

Java Exception Handling Lab Report

NullPointerException

Experiment	Null Pointer Exception Handling
Class Name	NullPointerExceptionDemo
Language	Java

1. What is NullPointerException?

NullPointerException (NPE) is a runtime exception in Java that occurs when a program tries to use a reference variable that currently contains **null**. In this program, **name** is assigned null, so calling **name.length()** tries to access a method through a null reference. The exception is caught by the catch block and a friendly message is displayed.

2. Objective

To understand NullPointerException in Java and demonstrate how to handle it using try-catch exception handling.

3. Task Description

Write a Java program in which a String reference is assigned the value null. Attempt to access its length, observe the NullPointerException, and handle the exception using a catch block so that the program terminates normally with an appropriate message.

4. Step-wise Execution

1. Start the program.
2. Enter the main() method.
3. Create the String reference variable name and assign null to it.
4. Execute name.length(). Since name does not refer to a String object, Java throws NullPointerException.
5. Control moves from the try block to the matching catch block.
6. The catch block prints: "Exception: Object contains null value".
7. The program ends normally.

5. Algorithm

START

Declare a String variable name and assign null.

Enter the try block.

Call name.length().

If name is null, NullPointerException occurs.

Catch NullPointerException.

Display the exception-handling message.

STOP

6. UML Diagram

NullPointerExceptionDemo
+ main(args: String[]): void try String name = null name.length() catch NullPointerException print message

7. Program Code

```
class NullPointerException {
    public static void main(String[] args) {
        try {
            String name = null;
            System.out.println(name.length());
        } catch (NullPointerException e) {
            System.out.println("Exception: Object contains null value");
        }
    }
}
```

8. Code Explanation

class NullPointerException — Defines the class named NullPointerException.

public static void main(String[] args) — The main method is the starting point of the Java program.

try { ... } — Contains statements that may produce an exception.

String name = null; — Creates a String reference named name but it does not refer to any String object.

name.length(); — Attempts to call length() through name. Because name is null, Java throws NullPointerException.

catch (NullPointerException e) — Catches the specific NullPointerException so the program can handle it.

System.out.println(...) — Displays a meaningful message instead of allowing the exception to terminate the program unexpectedly.

9. Output

Exception: Object contains null value

Output Screenshot

Java Program Output

Exception: Object contains null value

10. Result

The program successfully demonstrated NullPointerException and handled it using a try-catch block. The expected exception message was displayed without abnormal program termination.

11. Key Point

A null reference does not point to an object. Before calling methods or accessing members through a reference, make sure the reference is not null. For example, a null check such as **if (name != null)** can prevent this exception in appropriate situations.